

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (EE)

April/May 2012

EE 305 Power Electronics & Drives - I

Date: 04.05.2012, Friday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 Explain Principle of Phase control for Phase Controlled Rectifiers with circuit diagram and waveforms. [07]
- Q - 2 (a) Write down Advantages and Disadvantages of Power-electronic Converters. [03]
(b) Draw and explain IGBT switching characteristics. [05]
(c) Explain Resistance firing circuit for Thyristor (SCR). [06]

OR

- Q - 2 (a) A single-phase transformer, with secondary voltage of 230 V, 50 Hz, delivers power to load $R=10\Omega$ through a half-wave controlled rectifier circuit. For a firing angle delay of 60° , determine (a) the rectification efficiency (b) form factor (c) voltage ripple factor (d) transformer utilization factor (e) PIV of thyristor. [08]
(b) State and explain the various control strategies for varying duty cycle in Chopper. [06]
- Q - 3 (a) Draw and explain Two-quadrant Type-B Chopper with waveforms. [07]
(b) A dc series motor is fed from 600 V dc source through a chopper. The dc motor has the following parameters.
 $r_a = 0.04\Omega$, $r_s = 0.06\Omega$, $k = 4 \times 10^{-3} \text{ Nm / amp}^2$
The average armature current of 300 A is ripple free. For a chopper duty cycle of 60 % determine: (i) input power from the source (ii) motor speed (iii) motor torque [03]
(c) A step-up chopper has input voltage of 220 V and output voltage of 660 V. If the conducting time of thyristor-chopper is $100\mu\text{s}$, compute the pulse width of output voltage. In case output-voltage pulse width is halved for constant frequency, find the average value of new output voltage. [04]

OR

- Q - 3 (a) Draw and explain Static V-I characteristics of a Triac. And also Draw Cross sectional view of a Triac. [05]
- (b) Draw and explain I-V characteristics of a Light Activated SCR. [03]
- (c) Draw and explain firing circuit for a Triac using Diac with necessary waveforms. [06]

SECTION – II

- Q - 4 Explain Chopper control of DC series motor drive. [07]
- Q - 5 (a) Explain Three-phase Half-wave converter with R load with waveforms. [09]
- (b) Explain Line commutation of SCR with necessary waveforms. [05]

OR

- Q - 5 (a) Explain different speed control methods for a DC motor. [07]
- (b) Explain Current-limit control strategy for varying duty cycle in Chopper. [04]
- (c) A single-phase 230 V, 1 kW heater is connected across 1-phase, 230 V, 50 Hz supply through an SCR. For firing angle of 45° and 90° , calculate the power absorbed in the heater element. [03]

- Q - 6 (a) Explain single-phase Full-wave Mid-point converter with proper circuit diagram and waveforms. [08]
- (b) Explain Single-phase Full converter drive for a separately excited DC motor. [06]

OR

- Q - 6 (a) Draw and explain Resonant-Pulse (Class B) commutation. [07]
- (b) For the basic dc to dc converter of Figure 1, express the following variables as functions of V_s , R and duty cycle α in case load is resistive. [07]

- (a) Average output voltage and current (b) Output current at the instant of commutation. (c) Average and rms freewheeling diode currents (d) rms value of the output voltage (e) rms and average thyristor currents (f) effective input resistance of the chopper.

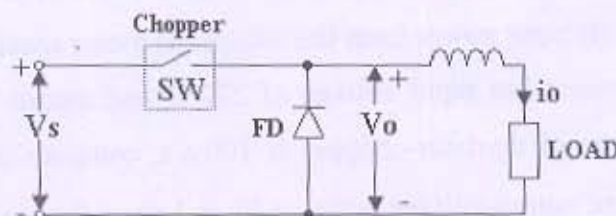


Figure 1
